

GAYATRI KHARCHE

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EDUCATION

UNIVERSITY OF CALIFORNIA, SAN DIEGO

San Diego, CA

Master of Science in Machine Learning and Data Science

Sept 2023 - May 2025

Fall 2023 Coursework: Statistical Learning, Probability and Statistics for Data Science, Programming for Data Analysis

Winter 2024 Coursework: Linear Algebra, Machine Learning Algorithms, Introduction to Deep Learning

Spring 2024 Coursework: Introduction to Visual Learning, Big Network Data, Random Processes

SAVITRIBAI PHULE PUNE UNIVERSITY

Pune, India

Bachelor of Engineering in Electronics and Telecommunication

Aug 2019 - May 2023

GPA: 4.00/4.00

SKILLS

Technical Skills: Python, SQL, MATLAB, C++, PyTorch, Tensorflow, Pandas, Numpy, Seaborn, Matplotlib, Scikit

Technologies: Data Science, Data Analysis, Data Visualisation, Machine Learning, Deep Learning, Cloud Computing, Artificial Intelligence, Internet of Things, Windows Server, Keras, LLM, Git, API, HTCondor, NLP

WORK EXPERIENCE

SAN DIEGO SUPERCOMPUTER CENTER

San Diego, USA

Machine Learning Intern

July 2024 – Present

- Conducted performance benchmarking of machine learning models using PyTorch on OSPool and utilized HTCondor to efficiently submit and manage computational jobs.
- Tested and benchmarked GPU performance on low-dimensional time series modeling tasks using CopulaCPTS and DeepSTPP frameworks.

HEWLETT PACKARD ENTERPRISE

Pune, India

Intern

Feb 2023 – Aug 2023

- Received intensive training in HPE GreenLake, hardware, network, cloud, server, VMware, and Linux fundamentals. Contributed to a VMware project showcasing expertise in domain establishment, virtual machine configuration, and implementation of DNS, NTP, and vCenter.
- Improved virtual environment performance, resulting in a 10% increase in operational efficiency. Proficient in using templates, content libraries, and migration strategies for streamlined virtualized environment management.
- Designed a scalable virtualized environment for testing and development, gaining hands-on experience in Windows Server Administration under Managed Services.

ORBITAL HOUSE PVT. LTD.

Pune, India

Project Intern

Dec 2021 – Jan 2022

- Played a pivotal role in a Home Automation project, utilizing ESP8266, cloud technologies, and various platforms.
- Integrated data analytics features to optimize energy usage, enhance user experience, and provide insights.

UNIVERSITY PROJECTS

IMAGE INPAINTING WITH CONTEXT ENCODERS FOR CIFAR-10 DATASET

Jan 2024 - Present

- Implemented an unsupervised auto-encoder method, achieving a binary cross entropy loss of approximately 0.54 for accurate prediction of pixel context.
- Trained Context Encoders to predict missing image regions and explored methodologies for sharper results.
- Analyzed effectiveness of representations in pre-training CNNs, demonstrating utility in inpainting through CIFAR-10 experiments with promising outcomes.

COVID-19's IMPACT ON CRIME RATES IN LA

Sept 2023-Dec 2023

- Conducted an in-depth analysis of Los Angeles crime rates before (pre-2019) and after (post-2020) the onset of COVID-19. Investigated the pandemic's impact on property crimes, violent crimes, and public order offenses (141 types).
- Utilized advanced data science techniques to uncover trends, patterns, and anomalies, providing insights into COVID-19's effects on community safety.

AUGMENTED REALITY ENABLED INTERNET OF THINGS FOR SMART AND INTERACTIVE ENVIRONMENT

September 2022 -August 2023

- Established an augmented reality lab with 5-6 diverse projects, including distance calculation, water level sensing, and augmented business cards.